AI Assisted Coding (III Year) Assignment

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Lab 17 – AI for Data Processing: Data Cleaning and Preprocessing Scripts

Objective

- To clean raw datasets using Python and AI-assisted scripting.
- To handle missing values, remove duplicates, and standardize data.
- To encode categorical variables and normalize numerical data.
- To automate preprocessing tasks for efficient data analysis.
- To understand the importance of preprocessing in improving model performance.

Task 1 – Customer Data Cleaning

Aim

To clean and preprocess a customer shopping dataset.

Procedure

- Loaded dataset using Pandas.
- Handled missing values in Age, Income, and City columns.
- Removed duplicate records.
- Standardized city names to lowercase.
- Encoded categorical variables such as Gender and City.

Code

```
import pandas as pd

df = pd.read_csv("shopping_trends.csv")

df['Age'] = df['Age'].fillna(df['Age'].median())
df['Purchase Amount (USD)'] = df['Purchase Amount (USD)'].fillna(df['Purchase Amount
(USD)'].mean())
df['Location'] = df['Location'].fillna(df['Location'].mode()[0])

df = df.drop_duplicates()

df['Location'] = df['Location'].str.lower().str.strip()
```

df = pd.get_dummies(df, columns=['Gender', 'Location'], drop_first=True)

print(df.head())

Result

A cleaned dataset with no missing values, standardized text, and encoded categorical features.

```
import pandas as pd

# Load dataset
df = pd.read_csv("/content/shopping_trends_updated.csv")

# Handle missing values
df['Age'] = df['Age'].fillna(df['Age'].median())
df['Purchase Amount (USD)'] = df['Purchase Amount (USD)'].fillna(df['Purchase Amount (USD)'].mean())
df['Location'] = df['Location'].fillna(df['Location'].mode()[0])

# Remove duplicates
df = df.drop_duplicates()

# Standardize text (city/location)
df['Location'] = df['Location'].str.lower().str.strip()

# Encode categorical variables
df = pd.get_dummies(df, columns=['Gender', 'Location'], drop_first=True)

print(df.head())
```

```
   Customer ID  Age Item Purchased  Category  Purchase Amount (USD) Size  \
0            1   55        Blouse  Clothing                     53    L
1            2   19       Sweater  Clothing                     64    L
2            3   50         Jeans  Clothing                     73    S
3            4   21       Sandals  Footwear                     90    M
4            5   45        Blouse  Clothing                     49    M

       Color  Season  Review Rating Subscription Status  ...  \
0       Gray  Winter            3.1                 Yes  ...
1    Maroon  Winter            3.1                 Yes  ...
2    Maroon  Spring            3.1                 Yes  ...
3    Maroon  Spring            3.5                 Yes  ...
4  Turquoise  Spring            2.7                 Yes  ...

   Location_south dakota Location_tennessee Location_texas  Location_utah  \
0                  False             False         False          False
1                  False             False         False          False
2                  False             False         False          False
3                  False             False         False          False
4                  False             False         False          False

   Location_vermont Location_virginia  Location_washington  \
0             False            False                False
1             False            False                False
2             False            False                False
3             False            False                False
4             False            False                False

   Location_west virginia  Location_wisconsin  Location_wyoming
0                   False               False             False
1                   False               False             False
2                   False               False             False
3                   False               False             False
4                   False               False             False

[5 rows x 66 columns]
```

Task 2 – Sales Data Preprocessing

Aim

To preprocess sales transaction data and create useful features.

Procedure

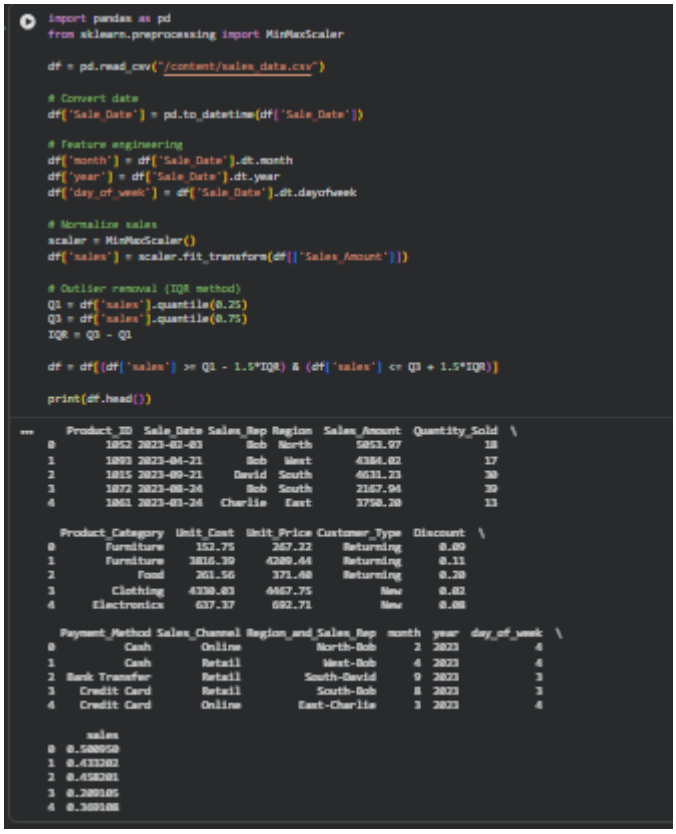
- Converted date column into datetime format.
- Extracted month, year, and day of the week.
- Normalized the sales column using Min-Max scaling.
- Detected and removed outliers using the IQR method.

Code

```
import pandas as pd
from sklearn.preprocessing import MinMaxScaler

df = pd.read_csv("sales_data.csv")

df['date'] = pd.to_datetime(df['date'])

df['month'] = df['date'].dt.month
df['year'] = df['date'].dt.year
df['day_of_week'] = df['date'].dt.dayofweek

scaler = MinMaxScaler()
df['sales'] = scaler.fit_transform(df[['sales']])

Q1 = df['sales'].quantile(0.25)
Q3 = df['sales'].quantile(0.75)
IQR = Q3 - Q1

df = df[(df['sales'] >= Q1 - 1.5*IQR) & (df['sales'] <= Q3 + 1.5*IQR)]

print(df.head())
```

Result

A transformed dataset with normalized values and additional time-based features.



```
import pandas as pd
from sklearn.preprocessing import MinMaxScaler

df = pd.read_csv("/content/sales_data.csv")

# Convert date
df['Sale_Date'] = pd.to_datetime(df['Sale_Date'])

# Feature engineering
df['month'] = df['Sale_Date'].dt.month
df['year'] = df['Sale_Date'].dt.year
df['day_of_week'] = df['Sale_Date'].dt.dayofweek

# Normalize sales
scaler = MinMaxScaler()
df['sales'] = scaler.fit_transform(df[['Sales_Amount']])

# Outlier removal (IQR method)
Q1 = df['sales'].quantile(0.25)
Q3 = df['sales'].quantile(0.75)
IQR = Q3 - Q1

df = df[(df['sales'] >= Q1 - 1.5*IQR) & (df['sales'] <= Q3 + 1.5*IQR)]

print(df.head())
```

```
   Product_ID  Sale_Date Sales_Rep Region  Sales_Amount  Quantity_Sold  \
0        1052 2023-02-03       Bob  North       5053.97             18
1        1093 2023-04-21       Bob   West       4384.02             17
2        1015 2023-09-21     David  South       4631.23             30
3        1072 2023-08-24       Bob  South       2167.94             39
4        1061 2023-03-24   Charlie   East       3750.20             13

  Product_Category  Unit_Cost  Unit_Price Customer_Type  Discount  \
0        Furniture     152.75     267.22     Returning      0.09
1        Furniture    3816.30    4280.44     Returning      0.11
2             Food     261.56     371.40     Returning      0.20
3         Clothing    4330.03    4467.75           New      0.02
4      Electronics     637.37     692.71           New      0.08

  Payment_Method Sales_Channel Region_and_Sales_Rep  month  year  day_of_week  \
0           Cash        Online            North-Bob      2  2023            4
1           Cash        Retail             West-Bob      4  2023            4
2   Bank Transfer        Retail           South-David      9  2023            3
3    Credit Card        Retail             South-Bob      8  2023            3
4    Credit Card        Online          East-Charlie      3  2023            4

      sales
0  0.500950
1  0.433202
2  0.458201
3  0.209105
4  0.369108
```

Task 3 – Healthcare Data Preparation

Aim

To preprocess healthcare data for machine learning.

Procedure

- Filled missing values in blood pressure and cholesterol.
- Encoded categorical variables using Label Encoding.
- Scaled numerical features using Standard Scaling.
- Split dataset into training and testing sets.

Code

```
import pandas as pd
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.model_selection import train_test_split

df = pd.read_csv("healthcare_dataset.csv")

df['blood_pressure'] = df['blood_pressure'].fillna(df['blood_pressure'].mean())
df['cholesterol'] = df['cholesterol'].fillna(df['cholesterol'].mean())

le = LabelEncoder()
for col in df.select_dtypes(include='object').columns:
    df[col] = le.fit_transform(df[col])

scaler = StandardScaler()
df[['blood_pressure', 'cholesterol']] = scaler.fit_transform(df[['blood_pressure', 'cholesterol']])

X = df.drop('target', axis=1)
y = df['target']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

print(X_train.shape, X_test.shape)
```

Result

A fully preprocessed dataset split into training and testing sets, ready for model training.

```
import pandas as pd
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.model_selection import train_test_split

df = pd.read_csv("/content/dirty_v3_path.csv")

# Handle missing values
df['Blood Pressure'] = df['Blood Pressure'].fillna(df['Blood Pressure'].mean())
df['Cholesterol'] = df['Cholesterol'].fillna(df['Cholesterol'].mean())

# Encode categorical
le = LabelEncoder()
for col in df.select_dtypes(include='object').columns:
    df[col] = le.fit_transform(df[col])

# Scale numerical features
scaler = StandardScaler()
num_cols = ['Blood Pressure', 'Cholesterol']
df[num_cols] = scaler.fit_transform(df[num_cols])

# Split data
X = df.drop('Medical Condition', axis=1)
y = df['Medical Condition']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

print(X_train.shape, X_test.shape)

(24000, 19) (6000, 19)
```

Task 4 – Employee Salary Data Preprocessing

Aim

To clean and preprocess employee salary data for analysis.

Procedure

- Handled missing values using forward fill.
- Removed salary outliers using the IQR method.
- Standardized department names to lowercase.
- Encoded job location and department using Label Encoding.

Code

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder

df = pd.read_csv("employee_salary.csv")

df = df.fillna(method='ffill')

Q1 = df['salary'].quantile(0.25)
Q3 = df['salary'].quantile(0.75)
IQR = Q3 - Q1

df = df[(df['salary'] >= Q1 - 1.5*IQR) & (df['salary'] <= Q3 + 1.5*IQR)]

df['department'] = df['department'].str.lower().str.strip()

le = LabelEncoder()
df['job_location'] = le.fit_transform(df['job_location'])
df['department'] = le.fit_transform(df['department'])

print(df.head())
```

Result

A structured dataset with cleaned and encoded features suitable for HR analytics.

```python
import pandas as pd
from sklearn.preprocessing import MinMaxScaler

df = pd.read_csv("/content/sales_data.csv")

# Convert date
df['Sale_Date'] = pd.to_datetime(df['Sale_Date'])

# Feature engineering
df['month'] = df['Sale_Date'].dt.month
df['year'] = df['Sale_Date'].dt.year
df['day_of_week'] = df['Sale_Date'].dt.dayofweek

# Normalize sales
scaler = MinMaxScaler()
df['sales'] = scaler.fit_transform(df[['Sales_Amount']])

# Outlier removal (IQR method)
Q1 = df['sales'].quantile(0.25)
Q3 = df['sales'].quantile(0.75)
IQR = Q3 - Q1

df = df[(df['sales'] >= Q1 - 1.5*IQR) & (df['sales'] <= Q3 + 1.5*IQR)]

print(df.head())
```

```
...   Product_ID  Sale_Date Sales_Rep Region  Sales_Amount  Quantity_Sold  \
0        1052 2023-02-03       Bob  North       5053.97             18
1        1093 2023-04-21       Bob   West       4384.02             17
2        1015 2023-09-21     David  South       4631.23             30
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  Product_Category  Unit_Cost  Unit_Price Customer_Type  Discount  \
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  Payment_Method Sales_Channel Region_and_Sales_Rep  month  year  day_of_week  \
0           Cash        Online            North-Bob      2  2023            4
1           Cash        Retail             West-Bob      4  2023            4
2  Bank Transfer        Retail           South-David      9  2023            3
3    Credit Card        Retail             South-Bob      8  2023            3
4    Credit Card        Online          East-Charlie      3  2023            4

      sales
0  0.580950
1  0.433202
2  0.458201
3  0.200105
4  0.369108
```

Conclusion

Data preprocessing is a crucial step in machine learning as it ensures data quality, consistency, and reliability. Proper handling of missing values, encoding, normalization, and outlier removal significantly improves model accuracy and performance.